

Student Guide — senior / module-05-dot-addition

IKS Curriculum

Table of Contents

Student Workbook

Student Workbook — Module 5 (Senior)

Name: _____ Class: ____ Section: ____

Day 1 — Baseline + Historicity

Baseline time (standard algorithm, 5×5 problem): _____ sec

The historicity question: was “Vedic mathematics” actually Vedic? Initial guess: _____

Day 2 — The 16 Sūtras

(Write them in your own hand here.)

1. ... (continue for all 16)

The two most relevant to addition: _____

Day 3 — Senior Dot Method

5×5 dot-method time: _____ sec

Compared to Day 1 baseline: _____ % faster/slower

Day 4 — Sūtra Apps

Worked: $234 + 98 = \underline{\hspace{1cm}}$ (Sankalana) Worked: $145 - 99 = \underline{\hspace{1cm}}$ (Sankalana) Worked: $99 \times 4 = \underline{\hspace{1cm}}$ (Ekanyūnena)

Day 5 — Base-n

In base 8: column 4, 7, 3, 6, 5 $\rightarrow$ units $\underline{\hspace{1cm}}$, carry $\underline{\hspace{1cm}}$ In base 16: column 8, A, 5, 7 $\rightarrow$ units $\underline{\hspace{1cm}}$, carry $\underline{\hspace{1cm}}$

Why does the dot method work in any base?

Day 6 — Dani Reading

Dani's three main critiques: 1.

$\underline{\hspace{10cm}}$ 2.
 $\underline{\hspace{10cm}}$ 3.
 $\underline{\hspace{10cm}}$

One sentence from the text that struck me:

My current position on the "Vedic mathematics" label:

Day 7 — Debate

My debate position: A / B / C

My 2 strongest arguments: 1.

$\underline{\hspace{10cm}}$ 2.
 $\underline{\hspace{10cm}}$

Strongest counter I steel-manned:

Days 8–9 — Paper

Track: **Math** / **History** (circle)

Working title:

Thesis paragraph:

3 sources I'll cite: 1. _____ 2. _____ 3. _____

Day 10 — Reflection

1. Track and topic chosen + why.

2. Most challenging part.

3. With 3 more days, what to research deeper.

Glossary

Sanskrit	Meaning
sūtra	aphorism
Bindu Paddhati	dot method
Sankalana Vyavakalanābhyām	by addition and by subtraction
Pūraṇāpūraṇābhyām	by completing or non-completing
Ekādhikena Pūrveṇa	by one more than the previous
Nikhilam Navataścaramaṁ Daśataḥ	all from 9, last from 10
Ūrdhva-Tiryagbhyām	vertically and crosswise
Pariśiṣṭa	appendix-text

Day 1 — Baseline + The Historicity Question

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Students record a personal baseline for standard-algorithm column addition AND consider the question: was “Vedic mathematics” actually Vedic?

Materials

- Baseline sheet: a single 5-addend × 5-digit column sum
- Stopwatch

- Printed extract: Tirthaji's preface to *Vedic Mathematics* (1965)

Flow

Warm-up (5 min)

Provocation: “You’ve all heard of ‘Vedic Mathematics.’ What does the ‘Vedic’ part mean?”

Take 3 student responses. Don’t correct. Note variety of views.

Baseline timing (5 min)

- 5-addend $\times$ 5-digit sum (e.g. $57283 + 14916 + 62385 + 27548 + 31057$).
- Solve using your standard method. Record time.

Core (25 min) — The 16 sūtras and the provenance puzzle

1. **The 16 sūtras** (write on board, abbreviated):

- | | |
|---|------------------------------------|
| 1. Ekādhikena Pūrveṇa | 2. Nikhilam Navataścaramaṁ Daśataḥ |
| 3. Ūrdhva-Tiryagbhyāṁ | 4. Parāvartya Yojayet |
| 5. Śūnyaṁ Sāmyasamuccaye | 6. (Ānurūpye) Śūnyam Anyat |
| 7. Sankalana Vyavakalanābhyāṁ (← we'll meet this one) | |
| 8. Pūraṇāpūraṇābhyāṁ (← and this one) | |
| 9. Calana-Kalanābhyāṁ | 10. Yāvadūnam |
| 11. Vyaṣṭi-Samaṣṭiḥ | 12. Śeṣānyaṅkena Caramena |
| 13. Sopāntya-dvayaṁ Antyaṁ | 14. Ekanyūnena Pūrveṇa |
| 15. Guṇitasamuccayaḥ | 16. Guṇaka Samuccayaḥ |

2. **Tirthaji's claim.** Read aloud the relevant sentence from Tirthaji's preface (in handout): > “The sūtras (aphorisms) of *Atharva-veda Pariśiṣṭa* are an integral part of the Vedas...”

(Tirthaji claims these 16 sūtras come from a Vedic appendix-text called the Atharva-veda Pariśiṣṭa.)

3. **The puzzle:** No other Sanskritist has ever produced the manuscript of this *Pariśiṣṭa*. Tirthaji died (1960) before producing it; his book was published posthumously (1965).
4. **What's at stake.** If the sūtras are Vedic, they're 3000+ years old — a major historical claim. If they're not, they're a brilliant 20th-century synthesis — *which would also be a real achievement, just a different one.*
5. **Preview Day 6.** “Day 6 we read the scholarly critique by S. G. Dani (1993). For now, hold the question open.”

Wrap-up (10 min)

- Quick survey on the board:
 - Hands up if you think the sūtras are likely Vedic.
 - Hands up if you think they're more likely 20th-century.
 - Hands up if you don't know yet.

- Discuss the distribution.
- Homework: read Tirthaji's preface (handout). Bring questions.

Differentiation

- **Tier up:** Read Plofker (2009), Ch. 1 — the broader history of Indian mathematics. What was happening in the 19th–early 20th c. that might explain Tirthaji's project?
- **Tier down:** Focus on memorising the 16 sūtra names this week. The provenance question deepens at Day 6.

Teacher notes

- Most students will start sympathetic to the “Vedic = ancient” claim. That's a starting point, not a fault.
- Don't decide the question for the class. The point is that they read primary and secondary sources, then form their own position.

Citation(s) used

- Tirthaji (1965), *Vedic Mathematics*, Preface.
- *Microsoft Word - sutras1.pdf* — the sūtra list.

Day 2 — The 16 Sūtras

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Students name all 16 Tirthaji sūtras + 13 sub-sūtras and gloss the meaning of at least 5.

Materials

- Printed table: all 16 + 13 sūtras with English translations
- *Microsoft Word - sutras1.pdf* reference

Flow

Warm-up (5 min)

- Recall from Day 1: how many sūtras? (16 + 13 sub-sūtras.)

Core (25 min)

1. Walk through the 16 sūtras with translations:

#	Sanskrit	English	Domain
1	Ekādhikena Pūrveṇa	“By one more than the previous”	Multiplication, division
2	Nikhilam Navataścaramaṇi Daśataḥ	“All from 9, last from 10”	Subtraction, multiplication

#	Sanskrit	English	Domain
3	Ūrdhva-Tiryagbhyām	“Vertically and crosswise”	General multiplication
4	Parāvartya Yojayet	“Transpose and apply”	Division
5	Śūnyam Sāmyasamuccaye	“When the sum is the same, that sum is zero”	Equations
6	(Ānurūpye) Śūnyam Anyat	“If one is in ratio, the other is zero”	Equations
7	Sankalana Vyavakalanābhyām	“By addition and by subtraction”	Equations, mental math
8	Pūraṇāpūraṇābhyām	“By completing or non-completing”	Equations, mental math
9	Calana-Kalanābhyām	“By differentiation and integration”	Calculus (modern interpretation)
10	Yāvadūnam	“Whatever the extent of its deficiency”	Squaring near a base
11	Vyaṣṭi-Samaṣṭiḥ	“Specific and general”	—
12	Śeṣānyaṅkena Caramena	“The remainders by the last digit”	Division
13	Sopāntya-dvayam Antyam	“The penultimate and the last”	—
14	Ekanyūnena Pūrveṇa	“By one less than the previous”	Multiplication by 9, 99, etc.
15	Guṇitasamuccayaḥ	“The product of the sum is the sum of the products”	—
16	Guṇaka Samuccayaḥ	“All the multipliers”	—

2. **The two sūtras most relevant to addition:**

- #7 *Sankalana Vyavakalanābhyām* — “by addition and by subtraction” — the basis of the “add 10, subtract 1” trick (Day 4).
- #8 *Pūraṇāpūraṇābhyām* — “by completing or non-completing” — the basis of “scan a column for completions to 10” (the dot method’s core insight).

3. **The dot method itself is NOT one of the 16 sūtras.** It is a pedagogical technique that uses the spirit of sūtras 7 and 8. (Important to be honest about this; it comes back on Day 6.)

Sūtra-recall quiz (10 min)

- 8 single-mark questions on the back of the worksheet:
 - Match 6 sūtras to their English meanings.
 - “Which sūtra means ‘vertically and crosswise’?”
 - “Which 2 sūtras are most relevant to addition?”

Wrap-up (5 min)

- Preview Day 3: “Tomorrow — full dot-method at Senior level. 5-digit $\times$ 5-addend, sub-60-second target.”

Differentiation

- **Tier up:** Pick a sūtra (say, *Ūrdhva-Tiryagbhyām*) and look up its multiplication application. Bring a worked example on Day 4.
- **Tier down:** Focus on memorising the names of sūtras 1, 2, 3, 7, 8 — the most commonly used.

Teacher notes

- The “calculus” mapping for sūtra 9 (*Calana-Kalanābhyām*) is Tirthaji’s later interpretation — Senior students may notice this is anachronistic. Acknowledge: “Tirthaji extends his sūtras’ interpretation; whether all these extensions are defensible is part of the Day 6 historicity debate.”

Citation(s) used

- Tirthaji (1965), sūtra list.
- *Microsoft Word - sutras1.pdf* in SutraGanita content.

Day 3 — Dot Method at Senior Level

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Students perform the dot method on 5-addend $\times$ 5-digit sums in under 90 seconds.

Materials

- Practice sheets (3 levels: warm-up 3×3 , target 4×4 , stretch 5×5)
- Stopwatch
- Whiteboard

Flow

Warm-up (5 min)

- Quick make-10 chorus (should be automatic).
- Recall: dot procedure (scan, dot on completion to 10, count dots as carry).

Core (15 min) — Procedure refinement at speed

1. Demo on board: a 5×5 problem. Solve at speed (45 sec target).
2. Identify three skills that distinguish speed-users from slow-users:
 - **Eye scan pattern** — top-to-bottom, then jump to next-in-column-that-completes (rather than always one-at-a-time).

- **Mental running total** — keep it in your head; don't recompute.
 - **Carry insertion** — write the carry as a tiny digit AT THE TOP of the next column before starting that column.
3. Worked example with these three skills called out explicitly.

Drill (20 min)

Three rounds, increasing difficulty:

- **Round 1 (5 min):** 5×3 -addend $\times$ 3-digit sums. Self-timed.
- **Round 2 (8 min):** 5×4 -addend $\times$ 4-digit sums. Self-timed.
- **Round 3 (7 min):** 3×5 -addend $\times$ 5-digit sums. Self-timed.

Record times for Round 3 — compare to Day 5's retest target.

Wrap-up (5 min)

- Show of hands by Round 3 average time: <60 sec / 60–90 sec / 90+ sec.
- Preview Day 4: “*Tomorrow — Sankalana Vyavakalanābhyām and Pūraṇāpūraṇābhyām with worked examples beyond just addition.*”

Differentiation

- **Tier up:** Try 6-addend $\times$ 6-digit. Time-target: under 120 sec.
- **Tier down:** Stay at Round 2 level until comfortable.

Teacher notes

- The “eye scan pattern” is the speed multiplier. Most students serial-scan. Teach the jump-pattern: see one completion, mark it, scan rest of column for the NEXT one without restarting.

Citation(s) used

- *Vedic Mathematics Batch1.pdf*, §5.4.

Day 4 — Sūtra Applications

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Students apply *Sankalana Vyavakalanābhyām* and *Pūraṇāpūraṇābhyām* to a range of mental-arithmetic problems beyond simple addition.

Materials

- Whiteboard
- Drill sheet (mixed sūtra applications)

Flow

Warm-up (5 min)

- Quick recall of sūtras 7 and 8.

Core (25 min) — Sūtra in action

1. **Sankalana Vyavakalanābhyām** — “by addition and by subtraction.”
 - Addition: $33 + 9 = 33 + 10 - 1 = 42$
 - Subtraction: $53 - 9 = 53 - 10 + 1 = 44$
 - Larger: $234 + 98 = 234 + 100 - 2 = 332$
 - Generalisation: when one operand is near a base, replace it with **base $\pm$ deficiency** and apply.
2. **Pūraṇāpūraṇābhyām** — “by completing or non-completing.”
 - Completion: $47 + 53 = 100$ (complete to 100, no carry mess).
 - Non-completion as a technique: when partial sums equal a base, you “complete” them mentally.
3. **Application to the dot method.** The dot itself IS a Pūraṇāpūraṇābhyām application — every dot = “I completed to 10.” Connect this to the dot method procedure explicitly.
4. **A 4-question drill (10 min):**
 - $67 + 38 = ?$ (use Sankalana)
 - $234 - 99 = ?$ (use Sankalana)
 - Compute $4+6+7+3+8+2$ by Pūraṇāpūraṇābhyām (find completions first).
 - $99 \times 4 = ?$ (apply Ekanyūnena Pūrveṇa — “one less than the previous”; $99 = 100 - 1$, so $99 \times 4 = 400 - 4 = 396$)

Formative Part A (10 min)

- quizzes/formative.md Part A: 5 sūtra-application questions.

Wrap-up (5 min)

- Preview Day 5: “*Tomorrow — does the dot method work in other bases? Let’s find out.*”

Differentiation

- **Tier up:** Pick *Ūrdhva-Tiryagbhyām* (vertically-and-crosswise multiplication). Self-teach from *Vedic Mathematics.pdf* §3.
- **Tier down:** Focus on Sankalana additions only.

Teacher notes

- Some students will love the named-sūtra framing; some will find it pretentious. Both responses are fine. The math is what matters.

Citation(s) used

- *Vedic Mathematics Batch1.pdf*, §5.5 (Sankalana applications).
- Tirthaji (1965), Chapter on Pūraṇāpūraṇābhyām.

Day 5 — Base-n Generalisation

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Students extend the dot method to base 8 and base 16, identify what changes and what stays the same, and complete the formative quiz.

Materials

- Whiteboard
- Base-conversion reference card
- Mixed-base practice sheets

Flow

Warm-up (5 min)

- Provocation: “Does the dot method work in base 8? Base 2?”
- Take 2 student guesses. Don’t resolve yet.

Core (20 min) — Generalisation

1. **In base 10:** the dot represents “10 used up” → 1 carried.
2. **In base 8:** the dot represents “8 used up” → 1 carried.
3. **In base 16:** the dot represents “16 used up” → 1 carried.
4. **What changes:** the *target sum* (8 or 16 instead of 10).
5. **What stays the same:** scan procedure; dot count = carry; units = leftover.
6. **Worked example in base 8:**

Column (base 8): 4, 7, 3, 6, 5

- Start total: 0

- +4 → 4

- +7 → 11₍₁₀₎ = 13₍₈₎. Wait – in base 8, 8 is "10". So 4+7 = 11₍₁₀₎, which is 1 "8" + 3, so DOT on 7, total = 3.

- +3 → 6

- +6 → 12₍₁₀₎ = 14₍₈₎. That's 1 "8" + 4. DOT on 6, total = 4.

- +5 → 9₍₁₀₎ = 11₍₈₎. That's 1 "8" + 1. DOT on 5, total = 1.

- Final units (base 8): 1. Carries: 3.

Check: $4+7+3+6+5 = 25_{(10)} = 31_{(8)}$. Units = 1, “tens” (i.e. 8s) = 3. ✓

7. The general rule:

In base n , mark a dot every time the running total reaches or exceeds n . Subtract n from the running total. Final running total = units. Dot count = “carry to next column.”

Practice (10 min)

3 problems: - Base 8: column of 6 single-digit (in base 8: 0–7) numbers. - Base 16: column of 4 single-digit (0–F) numbers. (Hex.) - Base 10: a real 4×4 multi-column for comparison.

Formative quiz Part B (8 min)

- 5 questions: mostly base-conversion and base- n dot method.

Wrap-up (2 min)

- Preview Day 6: “*Tomorrow — the historicity question. Was ‘Vedic mathematics’ actually Vedic?*”

Differentiation

- **Tier up:** Base 2 (binary) dot method. (Trivial: every column entry is 0 or 1; the dot marks whenever you have 2 ones.)
- **Tier down:** Stay in base 10; review the procedure.

Teacher notes

- The base- n extension is a beautiful “of course it works” moment for math-track students. Lean into it.
- For computer-science-track students, base 2 dot method = binary half-adder logic.

Citation(s) used

- (No new RAG/SutraGanita citations — this is original mathematical extension.)

Day 6 — The Historicity Question

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Students read Dani’s “Myths and Reality: On Vedic Mathematics” and identify Dani’s three main critiques of Tirthaji’s provenance claim.

Materials

- Printed copy of Dani (1993/2001) — full text or teacher-prepared synopsis
- Tirthaji’s preface (handout from Day 1)

- Quiet reading time

Flow

Warm-up (3 min)

- “Today: a scholarly reading. S. G. Dani is a respected mathematician at TIFR/IIT Bombay. He published this critique in 1993.”

Reading (20 min)

- 20 min silent reading of Dani (1993).
- As they read, students note in their workbook:
 - Dani’s three main critiques (in their own words).
 - One sentence from the text that struck them.
 - One question they want to ask.

Discussion (15 min)

1. **Class compiles Dani’s three main critiques** on the board:
 - No manuscript of the *Atharva-veda Pariśiṣṭa* containing these 16 sūtras has ever been produced.
 - The Sanskrit style of the sūtras (as Tirthaji presents them) is more recent than Vedic.
 - The mathematical techniques have 19th–early-20th-century European parallels (Trachtenberg etc.); calling them “Vedic” overstates the cultural specificity.
2. **Discuss:**
 - Are these critiques *fair*? Does Dani steel-man Tirthaji?
 - Does the absence of a manuscript = absence of the tradition?
 - If the sūtras are 20th-century, does that diminish their value?
3. **Frame the path forward:**
 - “The sūtras might NOT be Vedic AND might still be a real Indian intellectual achievement. The two are not in opposition. Tirthaji’s achievement is the *synthesis and codification*, even if the *attribution* doesn’t survive scrutiny.”

Reading response (7 min)

- Each student writes 1 paragraph (4–5 sentences): “What is your current best position on whether ‘Vedic mathematics’ is fairly named?”

Wrap-up

- Preview Day 7: “Tomorrow — structured debate. Pick your position.”

Differentiation

- **Tier up:** Read Plofker (2009) Ch. 1 too. Compare Plofker’s broader history with Dani’s specific critique.
- **Tier down:** Focus on Dani’s three critiques as written; the synthesis reasoning is the goal.

Teacher notes

- Some students will react defensively to Dani’s critique. Honour the reaction; don’t dismiss it. Ask: “*What is Dani’s strongest claim? What’s his weakest?*”
- Dani’s tone is firm but not dismissive. Model that tone.

Citation(s) used

- Dani, S. G. (1993). “Myths and Reality: On Vedic Mathematics.” *Frontline*. Reprinted in *Resonance* (October 2001).

Day 7 — Discussion: How Should We Frame “Vedic Mathematics”?

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Students articulate and defend a labelling position with at least one citation and one steel-manned counter.

Materials

- Day 6 reading response in hand
- Whiteboard

Flow

Warm-up (3 min)

- Three positions on the board:

Position	One-line claim
A — Defend “Vedic”	The label “Vedic mathematics” is appropriate; Tirthaji’s attribution should be defended.
B — Drop “Vedic”	The label is misleading; call it “Tirthaji’s 16 sūtras” or “Indian mental arithmetic.”
C — Keep “Vedic” with caveat	Use Tirthaji’s label but with a footnote noting the contested provenance.

Debate (30 min)

- 5 min: self-select into groups by position.
- 10 min: groups prep — 2 main arguments + 1 steel-manned counter to each opposing position + 1 citation to use.
- 12 min: structured debate (4 min per group).
- 3 min: open floor cross-questions.

Oral assessment (during debate)

Teacher uses same rubric as Module 2 Senior Day 7 — positions, evidence, counter-argument, listening.

Wrap-up (7 min)

- 3 student reflections: did anyone’s position shift?
- Note: *“You don’t have to commit to a position tonight. But by Day 8, you’ll choose a paper topic, and the position you defend in the paper might be different from the one you defended today. That’s OK.”*

Differentiation

- **Tier up:** Take a position you DON’T agree with. Defend it.
- **Tier down:** Use your Day 6 reading response as your speaking notes.

Teacher notes

- This debate is more politically loaded than the Module 2 debate. Some students’ families have strong feelings. Model respect.
- The “Position C” hybrid is often the strongest scholarly position — but defend it on its merits, not as a compromise.

Citation(s) used

- Tirthaji (1965), Preface.
- Dani (1993).
- Plofker (2009) optional.

Day 8 — Project Session 1: Choose Track + Outline

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Each student chooses Math track OR History track, drafts a one-paragraph thesis, and writes a working outline.

Materials

- Project brief
- Rubric
- Drafting materials

Flow

Warm-up (5 min)

- Hand out rubric. Note: tracks have different rubric weighting (see project brief).

Track choice + planning (15 min)

- Each student commits to a track:
 - **Math track:** Generalise the dot method to base- n and prove correctness. Optional extension: compare complexity with standard algorithm.
 - **History track:** Argue a position on the labelling / provenance of “Vedic mathematics.”
- Write down: working title + thesis paragraph + 3 sources (must include 1 from Tirthaji, 1 from Dani, 1 from another).

Outline drafting (15 min)

Standard 5-6 page paper outline: - Introduction with thesis - Section 1 - Section 2 - Section 3 - Counter-argument + response - Conclusion

Begin drafting (10 min)

Start the introduction. Don't worry about perfection.

Homework

- Continue drafting at home. Aim for 1.5–2 pages by tomorrow.

Differentiation

- **Tier up:** Math track — propose your own algorithm extension (e.g. to multiplication using the same dot logic).
- **Tier down:** History track — narrow to a single quote from Tirthaji and analyse it deeply.

Teacher notes

- Math-track students need access to LaTeX or clean number-formatted writing for proofs.
- History-track students need ALL three sources (Tirthaji preface, Dani, Plofker) at hand.

Day 9 — Project Session 2: Draft + Counter

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Each student reaches at least 2 drafted pages including a counter-argument and response.

Materials

- In-progress drafts

Flow

Warm-up (3 min)

- Brief: today's focus is the counter-argument.

Drafting + buddy work (35 min)

Sprint 1 (15 min) — Draft body. Quiet writing.

Sprint 2 (15 min) — Counter-argument workshop. - Buddy pair-up (match math + history when possible — fresh eyes). - Read each other's drafts. - Each pair gives the strongest counter to the other's thesis. - Student notes the counter and decides response.

Sprint 3 (5 min) — Incorporate. Each student writes the counter-argument paragraph or sketches its placement.

Wrap-up (7 min)

- 3 students share the best counter they discovered.
- Outline-to-date submitted.

Homework

- Finish complete draft. Bring tomorrow.

Differentiation

- **Tier up:** Two distinct counter-arguments — different angles.
- **Tier down:** Submit detailed outline + 1 paragraph if a full draft isn't ready.

Teacher notes

- Cross-track buddy pairing (math + history) often yields strong counters because the perspectives differ.

Day 10 — Summative + Paper Submission

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Students complete summative quiz and submit 5–6 page paper.

Materials

- Summative quiz (quizzes/summative.md)
- Stopwatch

Flow

Quiz (35 min)

- 35 marks. MC, sūtra identification, dot method + base- n , sūtra-application, short answer on historicity, open response.

Paper submission (10 min)

- Submit printed paper or upload.
- Teacher quick-checks: title page + thesis, page count, works-cited.

Wrap-up

- Reflection slip.

Homework

- None. Module complete.

Differentiation

- **Tier up:** Students who finish early start the next module's prerequisite reading.
- **Tier down:** Open-notebook quiz for documented learning needs.

Teacher notes

- Mark papers within 5 school days.
- Math-track papers: check the proof for correctness.
- History-track papers: check that Dani is steel-manned, not strawmanned.

Writing Prompts

Writing Prompts — Module 5 Dot Addition (Senior)

Reflection

1. **(Day 1)** Before this module, what did you assume about “Vedic mathematics”? Be specific about where the assumption came from.
2. **(Day 6)** After reading Dani — has your view shifted? In what specific way?

Explanatory

3. **(Day 3 or 5)** Explain the dot method procedure to a college-level audience, then prove its correctness in 1 page. Use place-value language.
4. **(Day 5)** Explain why the dot method works in any positional base. Use one specific example in a non-decimal base.

Argumentative

5. **(Day 6 or 7)** Argue: “The Vedic-provenance claim for Tirthaji’s sūtras is not central to their value.” 1–2 pages with citations.
6. **(Day 6)** Argue the OPPOSITE: “Calling the techniques ‘Vedic’ matters significantly — both for accuracy and for cultural recognition.” 1–2 pages.

Cross-disciplinary

7. (Math track) Compare Tirthaji's *Ūrdhva-Tiryagbhyām* multiplication with the standard long-multiplication algorithm. Complexity $O(n^2)$ for both — discuss the constant-factor differences.

8. (History track) Place Tirthaji's project (1925–1960 work, posthumous 1965 publication) in the context of the Indian independence movement and post-colonial scientific identity. 2 pages.

Grading

Same as Module 2 Senior — clear thesis, specific citations, fair counter-argument.

Homework

Homework — Module 5 (Senior)

Night	Task	Time
Day 1 → 2	Read Tirthaji's preface. Note 1 question.	20 min
Day 2 → 3	Memorise the 16 sūtras (Sanskrit names only).	20 min
Day 3 → 4	Practise 10 dot-method problems at home.	15 min
Day 4 → 5	Review sūtras 7 & 8 worked examples.	15 min
Day 5 → 6	Prep for Dani reading — review provenance question.	15 min
Day 6 → 7	Choose your debate position; prep 2-min argument.	30 min
Day 7 → 8	Submit project plan: track + thesis + 3 sources.	30 min
Day 8 → 9	Draft 1.5–2 pages.	60–90 min
Day 9 → 10	Complete draft; proofread; format.	60–90 min